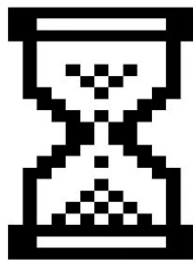


Starting Soon



Downtown Walk by | e s c p |
<https://www.escp.space>



Open Source Cyber Security: Crypto Crimes

👋 In chat, Introduce yourself with your **name**, **pronouns**, and **current obsession**



We're Recording

View it on Youtube

youtube.com/@Arachne_Digital/



Open Source Cyber Security: Crypto Crimes



ARACHNE

An abstract digital background on the left side of the slide. It features glowing blue and purple lines that form a complex, swirling pattern. Interspersed within these lines are various characters, including binary digits (0s and 1s) and some letters. The overall effect is a sense of data flow and digital connectivity.

We Will Cover

- Introductions & Icebreaker
- Crypto Crimes
- Questions & Answers
- Announcements



Arachne Digital

We advance global cyber security knowledge and empower organisations with bespoke consulting, our open-source tool, Thread, and access to our threat intelligence database.

Learn more at <https://arachne.digital/>

Hello

Vic Pe

They/Them



Community Manager
Arachne Digital

Kade Morton

He/They



Co-Founder
Arachne Digital

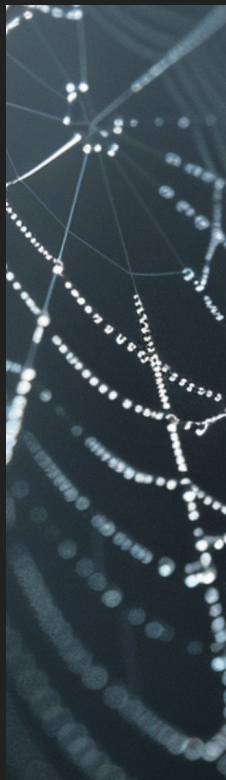
Jenine Carron

She/Her



Co-Founder
Arachne Digital

Previously



Community Call

Open Source Cyber Security: A Walkthrough of Thread

Learn how to download, use, and contribute to Thread, an open source cyber threat intelligence tool.

virtual

Where in the World are you?

✨ Austin, Texas, USA



☀️ Wellington, New Zealand

💬 In chat, share your **location**, **current time**, and an **emoji** you're vibing with

In *one* word.

How do you feel about Crypto?

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Up Next

- Introductions & Icebreaker
- **Crypto Crimes**
- Questions & Answers
- Announcements

What is cryptocurrency?

According to Merriam Webster:

Any form of currency that only exists digitally, that usually has no central issuing or regulating authority but instead uses a decentralized system to record transactions and manage the issuance of new units, and that relies on cryptography to prevent counterfeiting and fraudulent transactions

To break that down:

- Digital currency
- Usually not issued by an authority that is allowed to mint currency
- Usually decentralized
- Cryptography



Why is cryptography important?

There is more to it but the basics....

Public and Private Keys:

- You can generate two keys, a public key and a private key
- You make the public key publicly available
- People use that key to encrypt information
- Your private key is the only key that can reverse the encryption

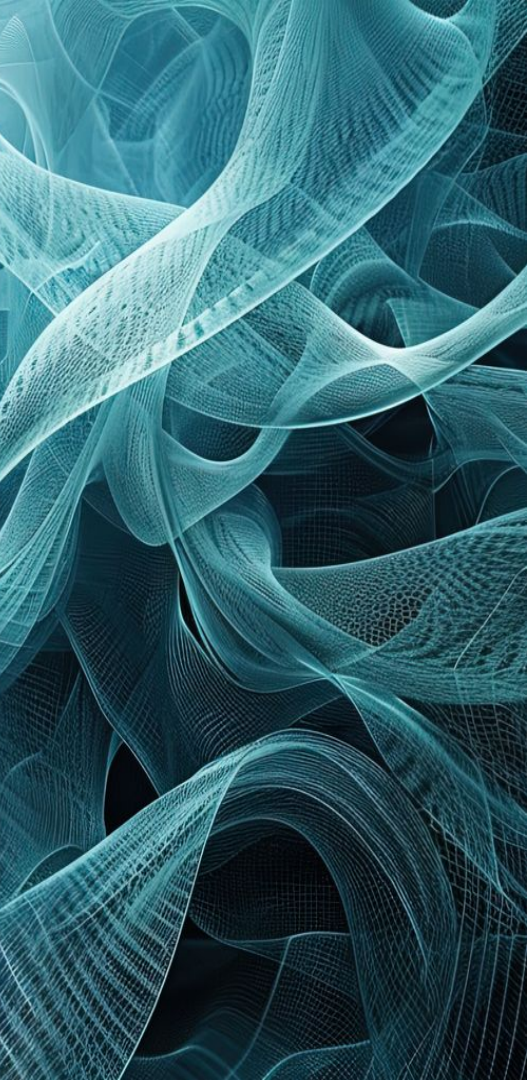
Why is cryptography important?

For cryptocurrency

- You sign the transaction with your private key,
only you have that key so it proves you approved the transaction
- The public key is used to prove that the private key used was your key
- This is stored in a ledger called a blockchain

This gets around problems such as ownership, repudiation, etc

This boils down to “cryptography provides trust.”



Why is cryptocurrency relevant to cyber security?

The international banking system has a lot of checks and balances built in to detect and stop criminality.

The cryptocurrency ecosystem has a lot fewer, *or* actively takes steps to support criminality.

As such, cryptocurrency facilitates *some* types of cybercrime.

Why is cryptocurrency relevant to geopolitics?

- Korea was one country under Japanese occupation at the end of World War 2 in 1945.
- The country was divided at the 38th parallel by the Soviet Union and the United States.
- Both Koreas formed their own governments in 1948 and disagreed with the other.
- On the 25th of June, 1950, the North invaded the South.
- UN forces, 90% of which supplied by the United States, invaded and retook the South.



Why is cryptocurrency relevant to geopolitics?

- With the South backed by the UN and the North backed by the Chinese and the Soviets, a war of attrition ensued, ending on the 27th of July, 1953.
- The end was an armistice, no peace treaty was signed. The Democratic People's Republic of Korea (North Korea) and the Republic of Korea (South Korea) are still at war.
- North Korea has a nuclear weapons program and is under heavy sanctions from most of the world.
- They have to fund their military somehow...

APT38

- Full aliases = TraderTraitor
- Partial Aliases = CryptoCore
- Organisation = Reconnaissance General Bureau (RGB)
- Country of organisation = Democratic People's Republic of Korea

Part of the North Korean military dedicated to espionage and stealing cryptocurrency to fund the regime.

This is one of many groups.



The scale of the problem

North Korea's government stole **\$1.34 billion worth** of cryptocurrency across **47 incidents** in 2024.

<https://www.chainalysis.com/blog/crypto-hacking-stolen-funds-2025/>

1
Jan
2025

NoOnes robbed for \$7.9 million

The breach led to a series of unauthorised withdrawals amounting to \$7.9 million, spanning the Ethereum, Tron, Solana, and Binance Smart Chain networks.

These funds were gradually siphoned out over two days via numerous small transactions, each typically withdrawing about \$7,000.

After being drained, the assets were sent to Tornado Cash for mixing.

No attribution to North Korea, but Tornado cash was used.

<https://www.crowdfundinsider.com/2025/01/235625-crypto-platform-noones-ceo-confirms-8m-hack-several-weeks-after-security-breach/>

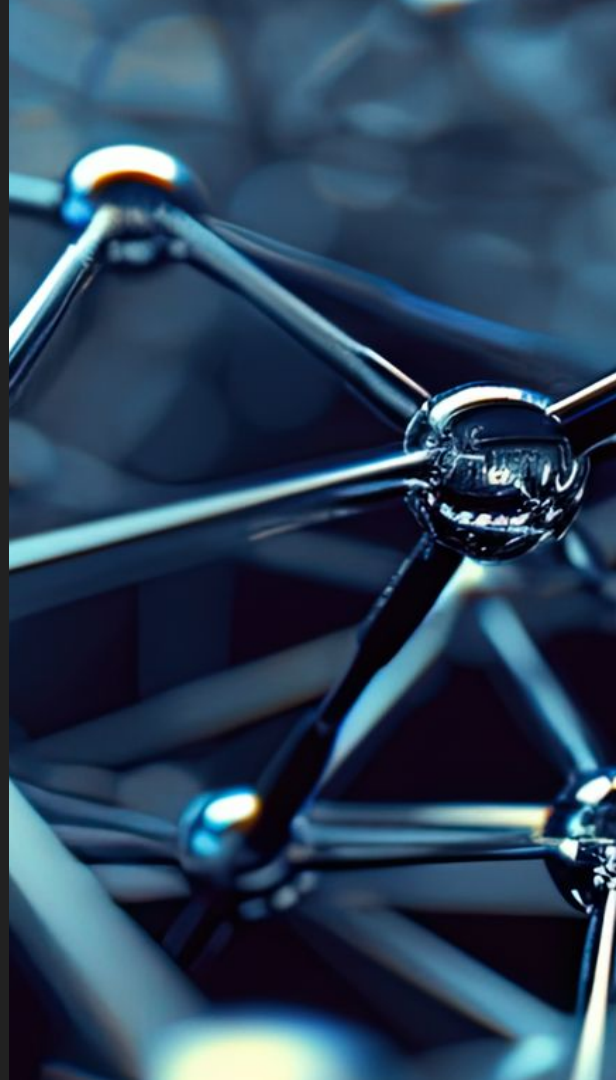
What's a *mixer*?

The blockchain, for each coin, keeps the transaction history. Stolen cryptocurrency is traceable.

A mixer is an exchange that has a pool of cryptocurrency. You put cryptocurrency in and you get new coins. These new coins have a history unrelated to the theft. The mixer takes a cut.

Tornado Cash is a mixer.

Zero knowledge mixers go one step further.



What are *Zero-Knowledge Proofs*?

In cryptography, **Zero-Knowledge Proofs enable users to prove they have certain information without revealing the information itself.**

You put cryptocurrency into a Zero Knowledge mixer, and it generates a note to prove your deposit but does not publicly tie back to the deposit.

To take coins out of the Zero Knowledge mixer, you present the note and create a zero-knowledge proof verifying you are owed coins, but is not linked to the deposit.

You get new coins, unrelated to the theft.

23
Jan
2025

Phemex Robbed for \$85 million

Singapore-based cryptocurrency platform Phemex was forced to pause some of its operations on Thursday after a cyberattack led to the theft of more than \$69 million in digital coins.

The digital assets were stolen through over 125 individual transactions recorded across 11 blockchain networks before the attackers started converting the funds into Ether via crypto mixing protocols like Tornado Cash.

Attributed to APT38

<https://therecord.media/69-million-stolen-cyberattack-crypto-platform-phemex>

<https://cointelegraph.com/news/lazarus-group-consolidates-bybit-phemex-hacker-wallet>

5
Feb
2025

Ionic Robbed for \$8.8 million

Ionic platform has been hacked, with hackers using unofficial fake LBTC (Lombard BTC) as collateral for lending on the Ionic platform. The estimated loss from this attack is about 8.8 million dollars.

No attribution and very little reporting.

<https://www.chaincatcher.com/en/article/2165720>

11
Feb
2025

zkLend Robbed for \$9.5 million

zkLend lost over \$9 million in a recent security incident. zkLend has asked the hacker to return 90% of the stolen funds. The firm promises not to take any legal action against the hacker if they return the funds and asked the hacker to keep the remaining 10% for their troubles.

It's ambiguous what happened as funds were transferred to a privacy pool called Railgun. It reverses transactions that come from theft.

There was little reporting on this.

No attribution and presumably North Korea wouldn't be silly enough to use Railgun?

<https://www.coinspeaker.com/zklend-seeks-return-8m-stolen-eth-security-breach/>

<https://u.today/vitalik-buterin-breaks-silence-on-zklend-exploit-flaunts-railgun>

19
Feb
2025

Bybit Hack Starts

The benign JavaScript file `app.safe.global` appears to have been replaced with malicious code on February 19, 2025, at 15:29:25 UTC specifically targeting the Multisig Cold Wallet of Bybit. The attack was designed to activate during the next Bybit transaction that occurred February 21, 2025, at 14:13:35 UTC.

Based on the investigation results from the machines of Bybit's signers and the cached malicious Javascript payload found on the Wayback Archive, we strongly conclude that AWS S3 or CloudFront account/API key of Safe.Global was likely leaked or compromised.

<https://thehackernews.com/2025/02/bybit-hack-traced-to-safewallet-supply.html>

20
Feb
2025

Bybit Hack Continues

Analysis from Silent Push has uncovered that the Lazarus Group (APT38) registered the domain `bybit-assessment[.]com` at 22:21:57 on February 20, 2025, a few hours before the cryptocurrency theft took place.

WHOIS records show that the domain was registered using the email address "trevorgreer9312@gmail[.]com," which has been previously identified as a persona used by the Lazarus Group in connection with another campaign dubbed Contagious Interview.

<https://thehackernews.com/2025/02/bybit-hack-traced-to-safewallet-supply.html>

21
Feb
2025

Bybit Robbed for \$1.4 billion

The cryptocurrency exchange Bybit was hacked for more than \$1.4 billion worth of Ethereum in what cybersecurity experts are calling the largest-ever theft targeting a cryptocurrency platform.

The transaction was manipulated through a sophisticated attack that masked the signing interface, displaying the correct address while altering the underlying smart contract logic.

Attributed to APT38

<https://therecord.media/hackers-drained-bybit-crypto-exchange-hack>

23
Feb
2025

Laundering

Within two hours of the theft, the stolen funds were sent to 51 different wallets, each holding approximately 10,000 ETH. These wallets were then emptied one by one over the next nine days.

The majority of the stolen Ether has now been converted to Bitcoin using eXch and other services. As with other North Korea-linked thefts, this bitcoin has now begun to be passed through mixers. This process has just begun, but already stolen assets worth hundreds of thousands of dollars have been sent through Cryptomixer and Wasabi Wallet.

<https://www.elliptic.co/blog/bybit-hack-largest-in-history>

26
Feb
2025

Laundering Continues

Within 48 hours, at least USD \$160 million had been funneled through illicit channels, with TRM estimating that the total surpassed USD \$200 million by February 23. By February 26, over USD \$400 million had been moved, indicating an unprecedented level of operational efficiency.

Initially, portions of the stolen Ethereum were routed through networks such as Binance Smart Chain and Solana, but the majority has now been converted directly into Bitcoin. Despite the swift movement of assets, most of the converted Bitcoin remains largely stationary, suggesting that the hackers are preparing for large-scale liquidation or further obfuscation through over-the-counter (OTC) networks.

<https://www.trmlabs.com/post/the-bybit-hack-following-north-koreas-largest-exploit>

28
Feb
2025

?? Robbed for \$3.1 million

An unknown victim is suspected of being hacked by Lazarus Group (APT38) on Tron for ~\$3.1M.

The funds were bridged from Tron to Ethereum and ETH was split between ten addresses before it was deposited to Tornado Cash.

The attacker reused a theft address from the Fantom exec hack in October 2023 which had been previously attributed to Lazarus Group by the UN.

<https://web-cdn.bsky.app/profile/zachxbt01.bsky.social/post/3ljcujujux4i22h>

3
Mar
2025

Laundering Continues

The initial 51 wallets the stolen cryptocurrency was moved to are now all drained.

<https://www.elliptic.co/blog/bybit-hack-largest-in-history>

4
Mar
2025

Laundering Continues

Nearly all of the robbed proceeds have been moved and converted into Bitcoin.

The funds are being held in ~4,400 addresses . The remaining ~10% of stolen funds have been lost to fees/freezes/off-ramped.

The second phase has entailed depositing an initial tranche of the stolen funds—now Bitcoin—into mixers.

Mixers usually receive a volume of a few million to \$10 million a day so, “whether these mixers can continue to absorb the amount of money at play is an open question.”

<https://techcrunch.com/2025/03/04/hackers-launder-most-of-bybits-stolen-crypto-worth-1-4-billion/>

5
Mar
2025

Laundering Continues

77% of the funds are still traceable.

<https://therecord.media/north-koreans-initial-laundering-bybit-hack>

12
Mar
2025

Laundering Continues

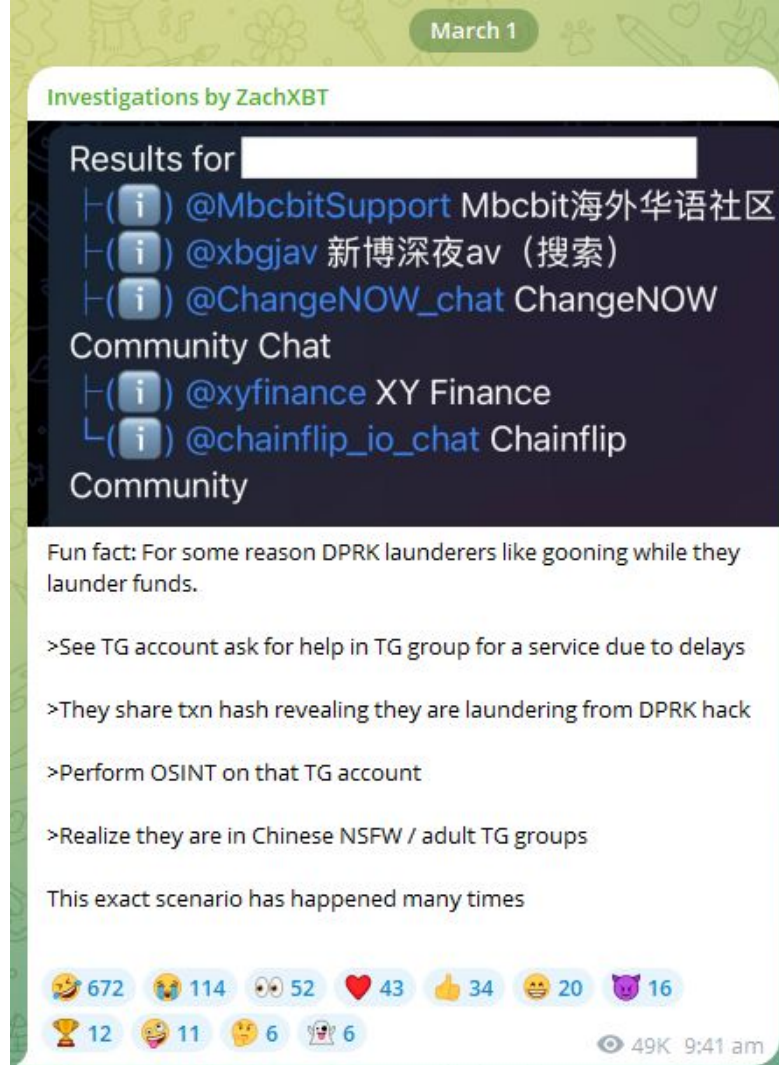
Lazarus Group (APT38) transfers \$750,000 from Bybit to Tornado Cash.

<https://news.bitcoin.com/north-koreas-lazarus-group-moves-400-eth-to-tornado-cash-following-1-5b-bybit-hack/>

How will they cash out?

Speculation that there are parties in China that will pay in cash for cryptocurrency, even though they know it's been stolen.

<https://t.me/investigations>





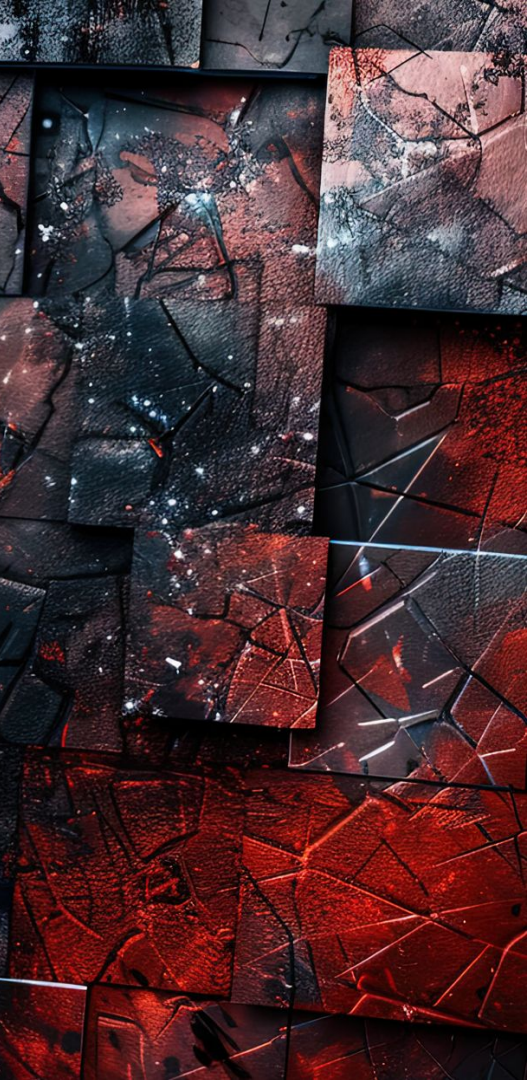
The new scale of the problem

North Korea's government stole **\$1.34 billion** worth of
cryptocurrency across **47 incidents** **in 2024.**

<https://www.chainalysis.com/blog/crypto-hacking-stolen-funds-2025/>

**Roughly \$1.5 billion stolen from 01 January 2025 to 17 March 2025
(when I stopped tracking).**

Some of this will be lost to the cost of laundering and some might be frozen by exchanges. But that is still a lot of money flowing to North Korea, potentially into their nuclear weapons program.



How does open source cybersecurity fit in?

There is a lot of cyber threat intelligence (CTI) in the public domain that can be used for research, especially related to cryptocurrency.

Tools like Thread can be used as part of a **Threat Informed Defence** program to help stem cryptocurrency funding North Korea's nuclear weapons program.

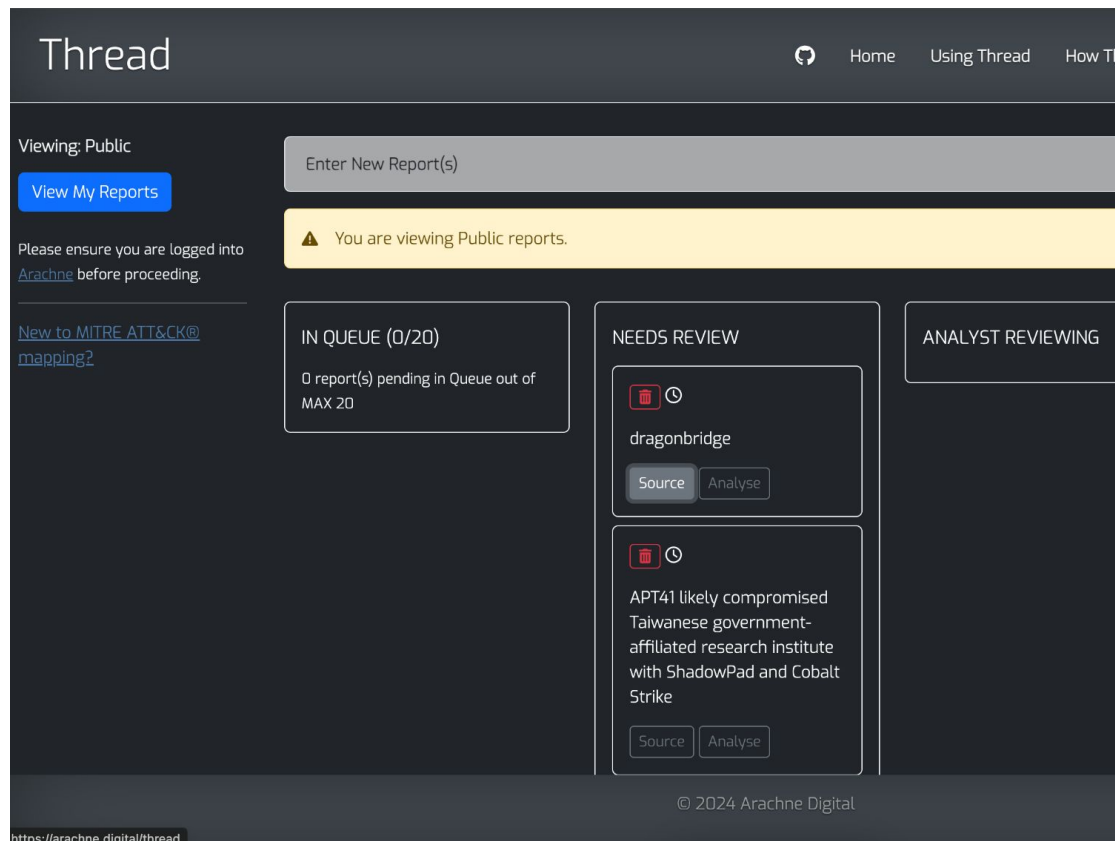
You can contribute to combatting this problem.

Thread

An open-source cyber security initiative dedicated to enhancing **cyber threat intelligence (CTI) mapping**.

It does this by automating the process of linking cyber security reports and articles to the MITRE ATT&CK® database for identification.

app.arachne.digital/thread





Up Next

- Introductions & Icebreaker
- Crypto Crimes
- **Questions & Answers**
- Announcements



Any Questions?

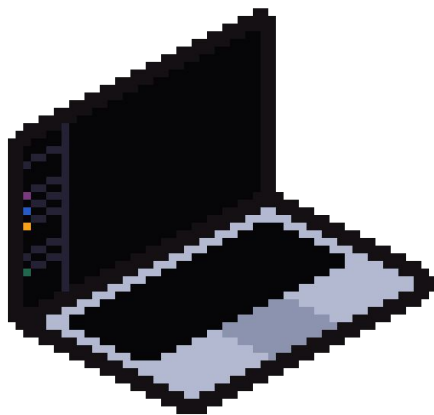


Up Next

- Introductions & Icebreaker
- Crypto Crimes
- Questions & Answers
- **Announcements**



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Next Week!



live virtual



Open Source Cyber Security Contributor Onboarding



Learn, connect, and contribute to the cyber security open source ecosystem with the Arachne Digital team



Continue the Conversation in Slack

<https://form.jotform.com/Arachne/request-for-slack-invitation>



Slack Invitation Request

Arachne Digital

Thank you for your interest in joining the Arachne Digital Slack! We look forward to collaborating with you on open source cyber threat intelligence projects. Please complete this quick questionnaire and we'll follow-up within 48 hours with an invitation to our Slack.



Thank you & Goodbye for Now!

The Thread Repository

github.com/arachne-threat-intel/thread

 **thread** Public

[forked from mitre-attack/tram](#)

Unwatch 3 Fork 3 Starred 4

main 2 Branches 6 Tags

Add file Code

This branch is 745 commits ahead of mitre-attack/tram:master Contribute Sync fork

 **dependabot[bot] and jecarr** Update nltk requirement from ~=3.8.2 to ~=... 5ff5775 · 2 months ago 841 Commits

 .github	Update run-tests.yml	4 months ago
 .idea	Updating the saving of the attack_dict.json file	5 months ago
 spindle @ aad79c1	Updated spindle	3 months ago
 tests	Pulling stuff apart to decouple them and make it easier to ...	3 months ago
 threadcomponents	Sentence-Tokeniser updates for nltk 3.8.2	2 months ago
 webapp	Updated JS dependencies	4 months ago
 .gitignore	Updating the .gitignore for virtual enviornments and for je...	6 months ago

About

Thread is a tool for analysts to map finished reports and articles to MITRE ATT&CK®.

[arachne.digital/thread](#)

cybersecurity threatintel hacktoberfest intelligence-analysis threat-intelligence mitre-attack cyber-threat-intelligence

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Arachne Digital Github Repos

github.com/arachne-threat-intel/



Arachne Digital

Providing timely and actionable cyber threat intelligence.

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Popular repositories

thread

Public

Forked from [mitre-attack/tram](#)

Thread is a tool for analysts to map finished reports and articles to MITRE ATT&CK®.

Python 4 3

tracery

Public

Forked from [searx/searx](#)

Privacy-respecting metasearch engine

Python 3

spindle

Public

community

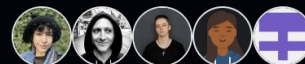
Public

👁 View as: Public ▾

You are viewing the README and pinned repositories as a public user.

You can [create a README file](#) visible to anyone.

People



Using Thread Online

app.arachne.digital/account/login/



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Username:

Password:

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Thread maps free text to MITRE ATT&CK an order of magnitude faster than doing it by hand

Why is mapping free text to ATT&CK useful?

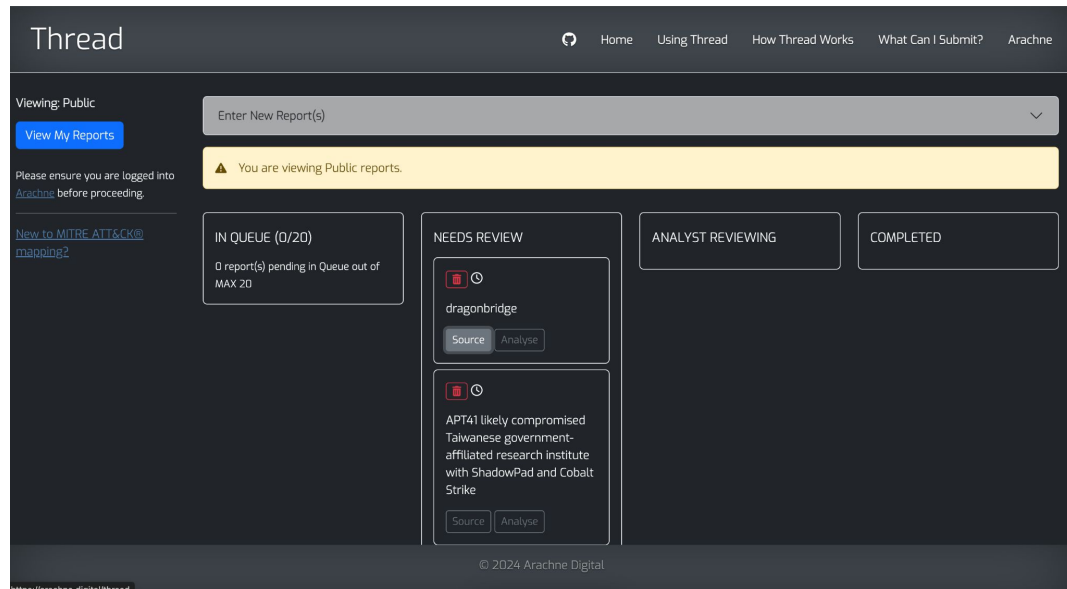
- Making free text structured for use in Thread Informed Defence
- Map incident reports to see the TTPs your SOC encounters
- Map threat intelligence reports etc

Thread

It is an open-source cyber security initiative dedicated to enhancing **cyber threat intelligence (CTI) mapping**.

It does this by automating the process of linking cyber security reports and articles to the MITRE ATT&CK® database for identification.

app.arachne.digital/thread





Viewing: Public

[View My Reports](#)

Please ensure you are logged into
[Arachne](#) before proceeding.

[New to MITRE ATT&CK@
mapping?](#)



⚠ You are viewing Public reports.

IN QUEUE (0/20)

0 report(s) pending in Queue out of
MAX 20

NEEDS REVIEW



dragonbridge

[Source](#)[Analyse](#)

APT41 likely compromised
Taiwanese government-
affiliated research institute
with ShadowPad and Cobalt
Strike

[Source](#)[Analyse](#)

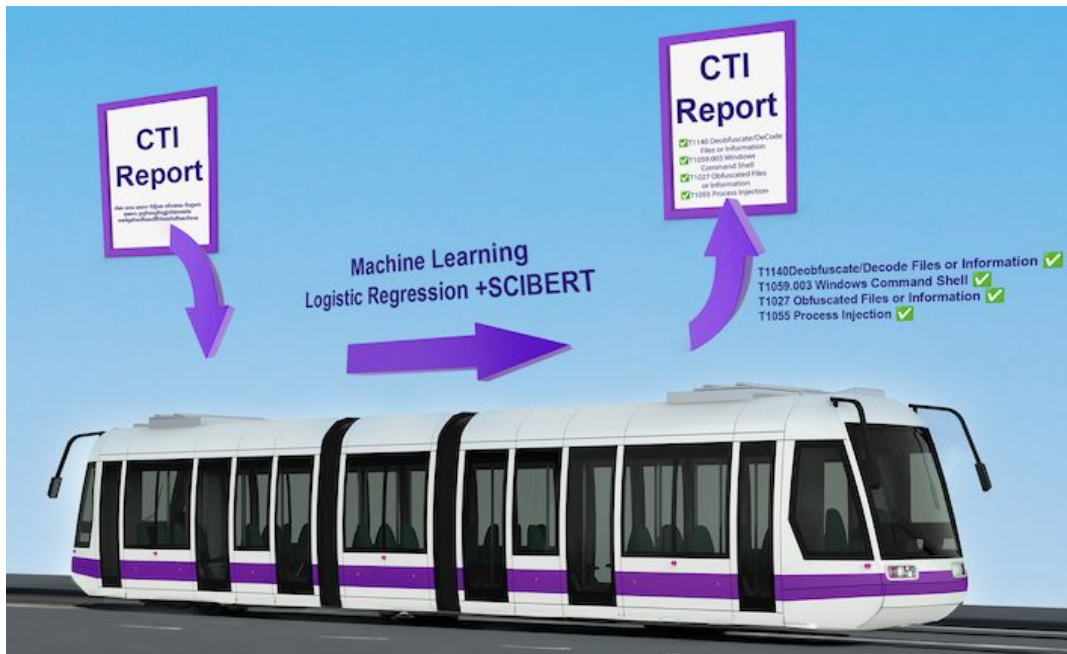
ANALYST REVIEWING

COMPLETED

Thread's Inception

Thread is a **MITRE TRAM** fork that is regularly maintained by Arachne Digital

<https://mitre-engenuity.org/cybersecurity/center-for-threat-informed-defense/our-work/threat-report-attck-mapper-tram/>



Any Questions?



Up Next

- 15 min | Introductions & Icebreaker
- 15 min | Thread Overview
- **10 min | Downloading Thread**
- 20 min | Questions & Answers